Dual-Branch Sparse Self-Learning With Instance Binding Augmentation for Adversarial Detection in Remote Sensing Images

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***Abstract*— Remote sensing image analysis technology based on neural networks has significantly facilitated human life. However, adversarial attacks can drastically impair the performance of these models, posing substantial economic and security risks. Current adversarial example (AE) detectors primarily focus on studying attacked natural images, while AEs in the remote sensing domain have not received adequate attention. To address this challenge, we propose a novel dual-branch sparse self-learning framework, leveraging instance binding augmentation. The contrastive branch concurrently enhances intra-instance and inter-example feature discrimination, while the masked branch reconstructs perturbation distributions. Furthermore, our method utilizes sparse encoding within depthwise separable convolutions to efficiently transfer parameters, thereby ensuring compatibility with deployment on mobile devices. Extensive experiments demonstrate that our method achieves state-of-the-art performance in detecting both white-box and black-box attacks on remote sensing images. Specifically, our method achieves an average detection accuracy of 95.69%/95.18%, a recall of 91.8%/93.6%, and an $F1$-score of 93.48%/94.22% on two attacked models across various attack scenarios, outperforming existing methods.**

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***Index Terms*— Adversarial examples (AEs), AE detection, instance binding augmentation, self-supervised, sparse depthwise separable convolution.**

I. Introduction

In RECENT years, significant advancements have been achieved in remote sensing image analysis technology, leveraging deep neural networks (DNNs) to unveil diverse features and dynamic changes on the Earth's surface [1], [2]. This has brought substantial convenience to various domains, including environmental monitoring [3], urban planning [4], disaster management [5], and agricultural development [6]. With the continuous improvement of DNNs and enhanced computational capabilities, these networks can identify and decipher complex patterns and structures within remote sensing images. DNNs display outstanding performance in tasks such as remote sensing image classification [7], object detection [8], and scene understanding [9] providing precise and reliable analytical results.

Nevertheless, research indicates the vulnerability of DNNs [10], [11], wherein even imperceptible changes in input data to the human eye can lead to entirely incorrect predictions. Concerns about the reliability of DNNs have been raised, particularly in the field of remote sensing images, with a notable focus on the existence of adversarial attacks [12], [13]. Adversarial attacks [14], [15] involve replacing clean images with carefully designed AEs, aiming to deceive specific visual models. As shown in Fig. 1, a maliciously attacked recognition model misclassifies a forest fire as a non-fire, missing the optimal time for fire control. In another example, a drone misclassifies a person as a tank, thereby reducing the vigilance of the observer. These adversarial attacks not only pose a threat to DNNs but also, when mishandling remote sensing image information, may lead to severe security consequences and substantial economic losses.

To address the problem of adversarial attacks, a significant amount of research has been focused on the design of AE detectors [16], [17]. AE detectors identify AEs and provide corresponding early warnings. Despite the increasing challenges posed by the diversity of attacks, there have been some effective detection algorithms. So far, two adversarial

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detection strategies exist. Supervised methods [18], [19] focus on identifying and understanding attack features by learning from labeled adversarial datasets. Unsupervised methods [20], [21], [22], on the other hand, aim to identify potential AEs by learning the intrinsic data structure of clean examples. As far as we know, current AE detectors mainly focus on studying attacked natural images, and AEs in the remote sensing domain have not received adequate attention. The spatial, spectral, and temporal complexities unique to remote sensing images, distinct from natural images, further increase the challenges of AE detection. Moreover, deploying AE detectors on mobile devices presents additional challenges.

In this article, our focus is on AE detection in remote sensing scenarios. We design a joint framework incorporating contrastive learning [23] and masked learning [24]. Distinct from traditional self-supervised learning methods [25], [26] that solely focus on the features within the sample itself and external to the sample. This article designs a novel dual-branch structure that takes into account the characteristics of adversarial attacks, where both clean and adversarial examples (AEs) coexist for the same image. The framework focuses on reconstructing the attack distribution and simultaneously learns the discriminative features between adversarial and clean examples. To implement the proposed framework, different from other methods that process one sample at a time, we propose a strategy that binds clean and AEs from the same sample to form twin instances. Within the twin instances, the same augmentation method is applied, while different augmentation pipelines are applied between instances. After passing through the encoder, the twin instances are separated during the decoding stage to execute the framework steps. On the other hand, considering the demand for deployment on mobile devices, we combine the idea of sparse coding with depthwise separable convolution and propose sparse depthwise separable convolution to reduce information loss. The module parameters can be directly transferred to the depthwise separable convolutions without further modification to the detection model. We create nine types of adversarial attack examples for detection on three remote sensing public datasets, including two attacked classification models. We assess the effectiveness and generalization of the proposed AE detection model through extensive experiments.

The main contributions of this article are summarized as follows.

1) We design a novel contrastive masked framework for AE detection, which applies a dual-branch structure to simultaneously reconstruct perturbation distributions and facilitate both intra-instance and inter-example feature discrimination.
2) We introduce a strategy of binding AEs with clean examples by adding twin augmentations, enabling the model to simultaneously extract salient features from both clean and adversarial instances to enhance the classification boundary.
3) To meet the deployment demands on mobile devices, we introduce a sparse depthwise separable convolution based on hierarchical hash coding, which can effectively reduce information loss without requiring modifications to the detection model.

The remainder of this article is organized as follows. Section II reviews the related work on adversarial attacks, self-supervised learning, and AE detection methods. Section III presents the proposed dual-branch sparse self-learning framework, detailing the instance binding augmentation and the sparse depthwise separable convolution. In Section IV, we describe the experimental setup used in this study and provide a comprehensive analysis of the experiments. Finally, Section V concludes the article with a summary and potential directions for future research.

II. Related Works

A. Adversarial Attack

AEs are usually generated by modifying clean samples with minimal cost to ensure that the AEs appear benign enough while causing misclassification in the model's output. Existing AE generation methods can be categorized into two types. In white-box attacks, the adversary has access to all information about the target model. In contrast, black-box attacks cannot access the target model and primarily rely on transferability. This problem can be defined as

$$\min_{x' \in \mathbb{R}^d} c(x, x') \; s.t. \; f_\theta(x') \neq y \tag{1}$$

where y is the true label associated with an example x being attack, the cost function c is $\| x' - x \|_p < \varepsilon$, p is Minkowski distance, and ε is the maximal allowed perturbation.

FGSM [27] is a one-step white-box attack where the clean example is moved along the gradient sign direction of the target model. BIM [28] is an iterative extension of FGSM, allowing for more refined adjustments to the examples. PGD [29] builds upon BIM by introducing random noise, making PGD more adept at evading the defense mechanisms. C&W [30] attack, as a representative work in the domain of white-box attacks, has demonstrated its ability to effectively deceive neural networks. The L_2 norm attack of C&W can be defined as

$$\begin{aligned} \boldsymbol{\varepsilon} = \text{minimize} & \left\| \frac{1}{2}(\tanh(w) + 1) - x \right\|_2^2 \\ & + c \cdot f\left(\frac{1}{2}(\tanh(w) + 1)\right) f(x + \boldsymbol{\varepsilon}) \\ & = \max\left(\max\left\{Z(x + \boldsymbol{\varepsilon})_{i:i \neq t}\right\} - Z(x + \boldsymbol{\varepsilon})_t, -\boldsymbol{\kappa}\right) \end{aligned} \tag{2}$$

where w is the target optimization variable, c is the regularization coefficient, Z is the example output vector, and $\boldsymbol{\kappa}$ is the penalty factor.

AOS [31] is a white-box attack method to enhance the transferability of targeted adversarial attacks through decision-level optimization. SSP [32] is a self-supervised white-box attack method that generates task-agnostic AEs by perturbing images with random noise and maximizing the perceptual feature distance.

In the domain of black-box attacks, various methods have been employed to deceive machine-learning models. ZOO [33] attack utilizes zero-order optimization to estimate gradients, while STA [34] employs spatial transformations such as rotation and translation to manipulate the output of the model.

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SA [35] deduces gradient information of input data by observing model changes. Sun et al. [36] propose a black-box attack in optical remote sensing images that optimizes and strategically places adversarial patches to significantly reduce the detection accuracy. ST data [37] presents three key attack strategies: enhancing inter-class similarity, proposing a model-specific loss function, and leveraging proxy data through a triple-player framework. These methods demonstrate the variety within the adversarial attack domain and highlight the challenges associated with maintaining model stability.

B. Self-Supervised Learning

Self-supervised learning has attracted widespread attention owing to feature quality and generalization capability, and many state-of-the-art models have been following this paradigm. Self-supervised learning is divided into two categories: contrastive learning and generative learning [38], [39].

The concept of contrastive learning involves pulling the features of similar data points closer while pushing those of dissimilar data points apart, requiring the model to distinguish between positive and negative instances. The classical discriminative contrastive learning method, SimCLR [40], employs two augmented versions of an image as positive examples, with the remaining images as negative examples. Subsequently, numerous contrastive learning methods have been improved based on SimCLR. Supervised contrastive learning [23] introduces the concept of labels and proposes the notion of class similarity. BYOL [41], on the other hand, introduces two neural networks and employs an exponential moving average (EMA) for gradient updates.

Generative learning involves reconstructing input data, with the model required to capture the intrinsic structure and contextual information of the data. MAE [24], renowned for its robust scalability, randomly masks parts of input images in the form of patches and then reconstructs the missing pixels. The VQ-VAE-2 [42] introduces a coding process involving quantization and hierarchical structures. PIRL [43] proposes invariant representation learning through pre-task objectives, thereby enhancing downstream task performance.

In real-world scenarios, the quantity of AE data is significantly smaller than that of clean examples. Furthermore, the lack of pre-trained weights for adversarial remote sensing images can adversely affect both the accuracy and convergence speed of the model. In this article, we propose a contrastive masked self-supervised learning method that simultaneously learns the boundaries between positive and negative examples and the internal structure of the data.

C. AE Detection

The AE detector is designed to reject AEs as input to third-party models, thereby reducing the risk of producing uncertain output results. Recent research on detection methods can be divided into two main categories: unsupervised detection and supervised detection [16], [44]. The objective of detectors is to construct a scoring function to differentiate AEs based on a threshold. This can be expressed as

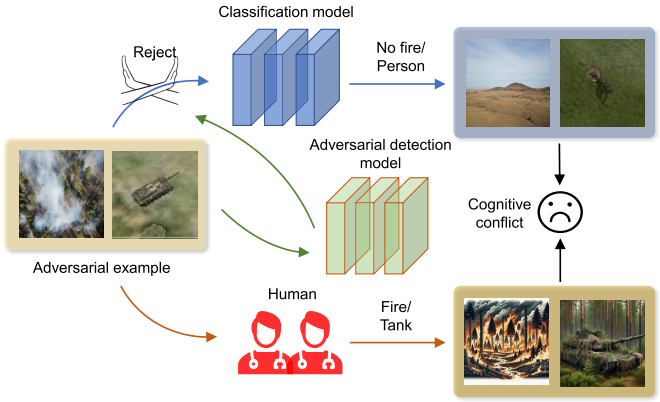


Fig. 1. Classification model in the illustration misclassifies AEs, while the AE detection model successfully prevents the input of AEs through a rejection strategy.

$$g(x) = \begin{cases} 1 & \text{if} \quad f(x) > \gamma \\ 0 & \text{if} \quad f(x) \leq \gamma \end{cases} \tag{3}$$

where $f(x)$ is an anomaly score encoder, $g(x) = 1$ is considered as AE, $g(x) = 0$ is considered as clean example, and γ is anomaly threshold.

Supervised methods simultaneously leverage adversarial and clean examples to distinguish between them using techniques such as logistic regression or classifiers. In the early stages of supervised learning, simplistic statistical methods are employed to compute the statistical characteristics of clean and AEs. NSS [19] computes 18 features from asymmetric generalized distribution and subsequently establishes an SVM classifier. With the advent of deep learning applications, some methods have been designed for end-to-end detection based on neural networks. PSGAN [45] employs GANs to generate AEs, enhancing the classifier's robustness by learning to differentiate between clean and AEs. ADS [46] employs an attention-based dual-stream method, demonstrating superior detection performance.

Unsupervised methods assume that the detector lacks prior knowledge about the attacker and can only rely on clean example data to reduce the available input feature space for adversaries. After obtaining a clean example representation, anomaly thresholds are applied to distinguish AEs. NIC [47] trains a one-class SVM classifier for each layer applying the layer's feature maps. AEAE [48] introduces the data depth-based method, which employs the logit layer to compute similarity scores concerning the training distribution.

However, existing research has primarily focused on AEs in natural images, lacking methods for detecting AEs in remote sensing images. In this article, we assume that the attacker's knowledge is available. In this situation, supervised learning methods are superior to unsupervised learning methods. Additionally, considering the implementation of the model on mobile devices, the detector is also considered for deployment on mobile platforms.

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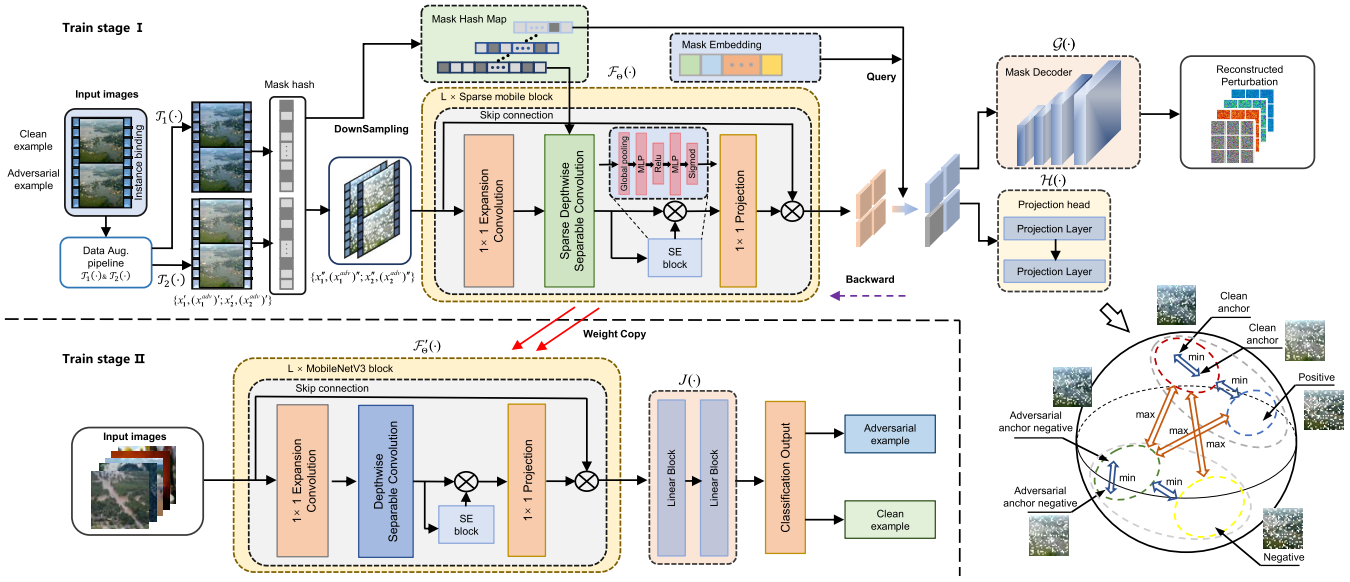


Fig. 2. Overview of the framework, including two train stages. In the first stage, self-supervised learning is conducted, incorporating contrastive masked learning based on the instance binding augmentation method. The second stage involves AE detection, transferring the parameters of sparse depthwise separable convolutions to downstream tasks.

III. Method

A. Contrastive Masked Framework

We propose a novel self-supervised learning framework, contrastive masked self-supervised learning, combining contrastive and generative learning. As shown in Fig. 2, the framework applies masks to the binding augmentation image instances and decodes features via two paths. One path predicts adversarial perturbation distributions for masked pairs, and the other discriminates clean and AEs. This dual-path framework enhances the model's capability to capture both inter-example and intra-instance representations, offering a focus on the reconstruction of adversarial perturbations. $L \times$ Sparse mobile block and $L \times$ MobileNetV3, respectively, indicate that the encoder is composed of L stacked sparse mobile blocks and L stacked MobileNetV3 blocks. In the self-supervised training stage, the encoder is composed of L sparse mobile blocks stacked together to extract features, while the decoder is divided into a decoder head and a projection head to learn an adversarial representation. In the detection training stage, the encoder is composed of L MobileNet-V3 blocks, and the decoder is a classification head consisting of two linear blocks.

The input image pairs consisting of clean and AEs are processed into two sub-image pairs by random data augmentation $\mathcal{T}_1(\cdot)$ and $\mathcal{T}_2(\cdot)$. Notably, the same data augmentation is applied to the same image pair, which can be represented as

$$\begin{aligned} \left\{x_1', \left(x_1^{adv}\right)'\right\} &= \mathcal{T}_1\left(x, x^{adv}\right) \\ \left\{x_2', \left(x_2^{adv}\right)'\right\} &= \mathcal{T}_2\left(x, x^{adv}\right) \end{aligned} \tag{4}$$

where (x, x^{adv}) denotes the clean and AE pair from the same origin image.

To further improve the representative ability of the model, we apply the random mask hashing strategy $\mathcal{M}_i(\cdot)\{i = 1, 2\}$, and employ hash downsampling maps combined with feature maps to reduce the loss of important features. Similar to data augmentation, the masks in the same image pair are identical. This can be expressed as

$$\begin{aligned} X_1 &= \left\{x_1'', \left(x_1^{adv}\right)''\right\} = \mathcal{M}_1\left(x_1', \left(x_1^{adv}\right)'\right) \\ X_2 &= \left\{x_2'', \left(x_2^{adv}\right)''\right\} = \mathcal{M}_2\left(x_2', \left(x_2^{adv}\right)'\right) \end{aligned} \tag{5}$$

where (X_1, X_2) denotes the images pair fed into the convolutional neural network $\mathcal{F}(\cdot)$.

Then, combined with decoder head $\mathcal{G}(\cdot)$ and projection head $\mathcal{H}(\cdot)$. The image pairs are fed into $\mathcal{F}(\cdot)$ to obtain the representation $Z_i = \mathcal{F}(X_i)$. $\mathcal{G}(\cdot)$ and $\mathcal{H}(\cdot)$ are employed for Z_i, respectively. The outputs can be denoted as

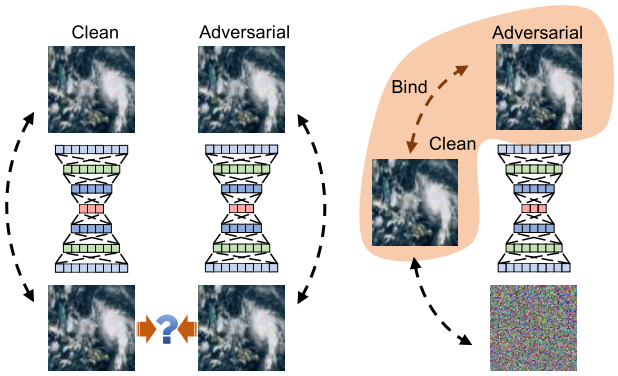
$$\begin{aligned} \left\{q_i, q_i^{\text{adv}}\right\} &= \mathcal{G}(Z_i) \\ \left\{k_i, k_i^{\text{adv}}\right\} &= \mathcal{H}(Z_i) \end{aligned} \tag{6}$$

where $i \in \{1, 2\}$ represents the two augmentation paths used in this study. We denote a sample within a given mini-batch by the index j, where $j \in \mathbb{Z}^+$. The decoding results of this sample are denoted as q_{ij}, and their projection representations are denoted as k_{ij}. We utilize q_{ij} to conduct masked learning to understand the distribution of adversarial attacks. On the other hand, the distribution of clean and AEs on the hypersphere is learned through performing adversarial contrastive learning between the projection representations k_{ij}.

B. Instance Binding Augmentation

In this section, we formally introduce the technique of instance-binding augmentation to enhance the representational capacity of self-supervised learning. As mentioned in the preceding section, we leverage image pairs, binding a clean example with its corresponding AE, to construct twin instances $\{x, x^{adv}\}$. Through the application of data augmentation $\mathcal{T}_i(\cdot)$ and masked $\mathcal{M}_i(\cdot)$, each twin instance contains two views.

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(a) Masked Autoencoders (b) Attack distribution prediction

Fig. 3. Two masked learning methods. (a) Original image reconstructed from MAE. (b) Attack distribution reconstructed employing binding augmentation strategy.

Subsequently, the two views are fed to the encoder $\mathcal{F}(\cdot)$, followed by decoder head $\mathcal{G}(\cdot)$ and projection head $\mathcal{H}(\cdot)$, to obtain the representation, respectively.

In previous masked learning papers, the design concept is to reconstruct the representation into the original image by applying the decoder. As shown in Fig. 3, however, such a methodology only captures the intrinsic structure of the data itself, leading to potential degradation in the AE of the decoder even if the prediction results are satisfactory. This is attributed to the high similarity between the distribution of clean examples and that of AEs, leading to a lack of distinctiveness in the output of the decoder. To address this issue, this article leverages twin instances of binding augmentation and directly predicts the perturbation distribution of adversarial attacks. This can be represented as follows:

$$\ell_{ij}^{\text{recon}} = \left\| q_{ij}^{adv} - \sqrt{\left(x_j - x_j^{adv}\right)^2} \right\|_2^2. \tag{7}$$

As clarified in contrastive learning, considering the influence within samples and sample classes, these contrastive methods ignore the similarity between clean examples and corresponding AEs from the same instance. Forcing all negative samples to the same embedding can lead to cognitive conflicts in the network, resulting in a decrease in representational capacity. The instance-binding augmentation method incorporates the internal relationship between diverse views of the same twin instance. The strategy maximizes the distance between clean examples and twin AEs while minimizing the dissimilarity between views. This can be represented as follows:

$$\ell_{m,n,j}^{\text{intra}} = -\log \frac{\exp\left(\text{sim}\left(k_{mj}, k_{nj}\right)/\tau\right)}{\sum_{n'=1}^{2N} \mathbb{1}_{[n' \neq m]} \exp\left(\text{sim}\left(k_{mj}, z_{nj}\right)/\tau\right)} \tag{8}$$

where $k_{mj}, k_{nj} \in \{k_1, k_1^{adv}, k_2, k_2^{adv}\}$, $\mathbb{1}_{[n' \neq m]}$ denotes an indicator function, τ is a temperature parameter, and $\text{sim}(k_{mj}, k_{nj}) = k_{mj}^{\top} k_{nj} / \|k_{mj}\| \|k_{nj}\|$ denotes cosine similarity.

C. Sparse Depthwise Separable Convolution

Masked reconstruction has achieved notable success in the domain of visual transformer self-supervision, extending to the CNNs presents challenges. In mobile computing, transformers struggle with convergence and low computational efficiency, while CNNs outperform in processing data of various resolutions and are more hardware-friendly. This stems from architectural differences between transformers and CNNs. Transformers divide the image into non-overlapping patches, and by simply masking specific patches, transformers can process other patches without impact. However, in CNNs, convolutions involve overlapping sliding windows, with significant shifts in data distribution caused by the masks, leading to information decay.

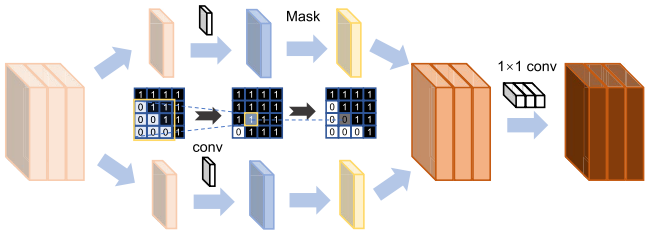


Fig. 4. Sparse depthwise separable convolutions correct data distribution destroyed by convolution.

Inspired by sparse coding, we employ sparse depthwise separable convolutions instead of depth-separable convolutions in the self-supervised learning stage as shown in Fig. 4. The method offers two advantages. First, it can reduce information loss during the downsampling stage of the masking process, ensuring that the information distribution remains unshifted and enhances the self-supervised representation ability. Second, while the proposed sparse coding module differs in structure from depthwise separable convolution, the two can share weights, and directly transferring parameters will not increase additional computational cost.

The mask hash values use one to indicate that the convolution is active (unmasked) in a region, while zero signifies that the region is masked, and the size of the mask hash aligns with the dimensions of the image. Subsequently, as convolutional networks change the size of the image during training, a corresponding adjustment must be made to the mask hash. Specifically, the mask hash is downsampled according to the image's size within each block, to ensure that it matches the new dimensions of the image. The computed mask hashes from multiple blocks are organized and stored in a mask hash map, where the retrieval operation for a specific mask hash is denoted by Hash$(\cdot)$. In the encoder, this Hash$(\cdot)$ operation is applied to locate and retrieve the corresponding mask hash for a given block from the mask hash map. Upon reaching a new block during the computation, we reset the positions indicated in the mask hash to zero in order to alleviate the distribution shifts caused by the convolutional operations. Encoder $\mathcal{F}(\cdot)$ contains L blocks, the output feature is $(\mathcal{S}_1 : H_1 \times W_1, \mathcal{S}_2 : H_1 \times W_1, \ldots, \mathcal{S}_L : H_L \times W_L)$, and the size is reduced successively. The sparse depthwise separable convolutions can be represented as

$$\mathcal{O}_i = (\mathcal{S}_i * \mathcal{K}_{di} * \text{Hash}(H_i, W_i)) * \mathcal{K}_{pi} \tag{9}$$

where $\mathcal{O}_i$ denotes block output, $\mathcal{K}_{di}$ and $\mathcal{K}_{pi}$ are, respectively, depthwise convolution and pointwise convolution.

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In the decoder, similar to the encoder, we apply Hash(·) to locate the corresponding mask hash in the mask hash map for a specific block. However, instead of setting the masked positions to zero to restore features, we generate learnable parameters referred to as mask embedding $\mathcal{M}_i$ at these masked positions. Through mask embedding, sparse features $\mathcal{S}_i$ are transformed into dense features $\mathcal{S}_i'$. We define the decoding convolutional block as $\mathcal{U}_i$, which applies $\mathcal{M}_i$ as the query to interact with surrounding features and restore the masked positions, as well as refine features at other positions. This can be represented as follows:

$$\mathcal{S}_i' = \mathcal{U}_i * (\mathcal{S}_i * \text{Hash}(H_i, W_i) + \mathcal{M}_i). \tag{10}$$

A projection layer $\mathcal{P}_i$ is employed to adjust the network width. The output from each block $\mathcal{B}_i$ is directly applied to the corresponding level of the decoder through skip connections for feature fusion, supplementing lower-level features. This can be represented as follows:

$$D_i = \mathcal{B}_i(D_{i+1}) + \mathcal{P}_i\left(S_i'\right). \tag{11}$$

During the adversarial detection train stage, masked maps are not applied, and all convolution values are activation values. The mask decoder is discarded, and the pre-trained sparse encoder parameters are directly transferred to depthwise separable convolutions.

D. Optimization

In this section, the overall optimization objective of the proposed method is introduced. We consider the contrastive loss to consist of two parts. Leveraging the instance binding augmentation strategy, the loss within instances is represented as $\ell_{m,n,j}^{\text{intra}}$. We employ a supervised contrastive loss to measure the distribution difference between clean and AEs

$$\ell_j^{\text{inter}} = -\frac{1}{|P(j)|}\sum_{p\in P(j)} \log\frac{\exp\left(k_j \cdot k_p/\tau\right)}{\sum_{a\in A(j)}\exp\left(k_j \cdot k_a/\tau\right)} \tag{12}$$

where $P(m)$ denotes the set of examples from the same class as j, excluding j itself, $|\cdot|$ denotes the number of elements in the set, and $A(m)$ denotes the set of all examples.

Given a mini-batch of (N) samples, we calculate the reconstruction loss, intra-contrastive loss, and inter-contrastive loss as follows:

$$\begin{aligned}
\mathcal{L}_{\text{recon}} &= \sum_{j=1}^{2N} \ell_{ij}^{\text{recon}} \\
\mathcal{L}_{\text{contra}}^{\text{inter}} &= \sum_{j=1}^{2N} \ell_j^{\text{inter}} \\
\mathcal{L}_{\text{contra}}^{\text{intra}} &= \frac{1}{2N}\sum_{j=1}^{N}\left(\ell_{2i-1,2i,j}^{\text{intra}} + \ell_{2i,2i-1,j}^{\text{intra}}\right)
\end{aligned} \tag{13}$$

where $\mathcal{L}_{\text{recon}}$ denotes reconstruction loss, $\mathcal{L}_{\text{contra}}^{\text{inter}}$ denotes inter-contrastive loss, and $\mathcal{L}_{\text{contra}}^{\text{intra}}$ denotes intra-contrastive loss.

The overall loss for network optimization is a combination of reconstruction loss, inter-contrastive loss, and intra-contrastive loss as follows:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \boldsymbol{\alpha}\mathcal{L}_{\text{contra}}^{\text{intra}} + \boldsymbol{\beta}\mathcal{L}_{\text{contra}}^{\text{intre}} \tag{14}$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are hyperparameters that balance the above-mentioned three losses.

IV. Experiments

We introduce the experimental datasets, attack pattern design, and experimental setup in Section IV-A. Furthermore, we report the results of the proposed model detection against white-box and black-box attacks in Section IV-B, and conduct an ablation study in Section IV-C to assess the effectiveness of individual components.

A. Experimental Setup

1) Datasets: We utilize remote sensing images to generate adversarial attack examples. Three popular datasets are considered: Cyclone Wildfire Flood Earthquake Database from Kaggle, RSSCN7 [49], and FLAME [50]. DWFED contains four categories of natural disasters, while the RSSCN7 comprises seven categories of remote sensing landforms. The FLAME dataset is a large-scale remote sensing dataset comprising 39 357 images with four categories of burning labels. Our objective is to manipulate the data distribution leads to incorrect classification results by the model.

2) Image Classifiers: By default, we consider two lightweight classification models suitable for real-time remote sensing scenarios. One of the image classifiers attacked employs a deep residual network with 18 hidden layers, referred to as ResNet18 [51]. We also adopt Mobilenet-V3 [52] as another image classifier that employs lightweight attention to improve classification performance.

3) Attack Models: This article evaluates the performance of the proposed detection method against other techniques under the same attacks with specific parameters, and the attack results are shown in Table I. Five latest white-box adversarial attack strategies are considered [27], [29], [30], [53], [54], whose hyperparameters are as follows: FGSM ($\epsilon = 128/256$), Deepfool (N/A), C&W (norm $= L_2$), PGD ($\epsilon = 128/256$, norm $= L_\infty$), and AutoAttack ($\epsilon = 128/256$, norm $= L_1$). Simultaneously, four black-box adversarial attack strategies are considered [33], [34], [35], [55], whose hyperparameters are as follows: ZOO (N/A), ST (N/A), SA ($\epsilon = 0.125$), and MoreBayesian (N/A). This article employs the ART toolbox and PyTorch to implement these adversarial attacks.

4) Performance Indicators: This article assesses the performance from the recall, precision, and $F1$-score. The recall, precision, and the $F1$-score are defined as follows:

$$\begin{aligned}
\text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\
\text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\
F1 &= 2\times\frac{\text{Recall}\times\text{Precision}}{\text{Recall}+\text{Precision}}
\end{aligned} \tag{15}$$

where TP denotes the number of correctly detected AEs, FN denotes the number of AEs that escape detection, and FP denotes the number of clean examples that are detected as AEs.

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TABLE I
CLASSIFICATION ACCURACY OF THE CLASSIFIERS SUBJECT VARIOUS ATTACKS

Dataset(Accuracy)	Model	No attack	FGSM	Deepfool	C&W	PGD	AutoAttack	SA	ST	ZOO	MoreBayesian
DWFED	Resnet18	**93.01%**	44.64%	17.81%	23.52%	7.56%	20.18%	36.75%	26.71%	45.51%	46.33%
	MobileNet-V3	**90.76%**	48.47%	23.11%	28.57%	8.00%	13.64%	28.85%	49.71%	52.98%	57.27%
RSSCN7	Resnet18	**87.14%**	44.48%	16.25%	27.5%	14.46%	12.5%	15.89%	26.07%	35.89%	34.46%
	MobileNet-V3	**81.07%**	45.17%	13.75%	35.85%	17.14%	4.82%	13.21%	24.82%	27.75%	31.25%
FLAME	Resnet18	**93.50%**	45.94%	15.57%	7.04%	6.49%	21.31%	10.15%	39.42%	43.96%	43.53%
	MobileNet-V3	**93.26%**	62.71%	32.89%	32.23%	6.73%	28.17%	19.14%	62.98%	65.76%	75.18%

TABLE II
PERFORMANCE ON DATASETS AND MODELS

Datasets	Attacks	ResNet18			MobileNet-V3		
		Precision	Recall	F1 score	Precision	Recall	F1 score
DWFES	C&W	97.37%	83.15%	89.70%	97.75%	97.75%	97.75%
	Deepfool	98.89%	100%	99.44%	100%	100%	100%
	FGSM	89.36%	94.38%	91.80%	90.91%	78.65%	84.34%
	PGD	95.38%	69.66%	80.52%	93.42%	79.78%	86.06%
	AutoAttack	80.51%	69.66%	74.69%	88.73%	70.78%	78.75%
	ST	100%	100%	100%	100%	100%	100%
	SA	100%	88.76%	94.05%	98.84%	95.51%	97.14%
	ZOO	98.89%	100%	99.44%	100%	100%	100%
	Morebayesian	100%	98.87%	99.43%	98.88%	100%	99.44%
RSSCN7	C&W	90.48%	67.86%	77.55%	74.55%	73.21%	73.87%
	Deepfool	96.55%	100%	98.25%	94.92%	100%	97.39%
	FGSM	89.13%	73.21%	80.39%	86.54%	80.36%	83.33%
	PGD	91.67%	78.57%	84.62%	89.13%	73.21%	80.39%
	AutoAttack	69.84%	78.57%	73.94%	68.49%	89.28%	77.51%
	ST	96.55%	100%	98.25%	96.55%	100%	98.25%
	SA	94.55%	92.86%	93.69%	96.36%	94.64%	95.50%
	ZOO	96.55%	100%	98.25%	94.92%	100%	97.39%
	Morebayesian	97.95%	85.71%	91.42%	100%	98.21%	99.09%
FLAME	C&W	100%	97.5%	98.73%	100%	100%	100%
	Deepfool	100%	100%	100%	100%	100%	100%
	FGSM	100%	100%	100%	100%	100%	100%
	PGD	100%	100%	100%	100%	100%	100%
	AutoAttack	100%	100%	100%	100%	100%	100%
	ST	100%	100%	100%	100%	100%	100%
	SA	100%	100%	100%	100%	95.83%	97.87%
	ZOO	100%	100%	100%	100%	100%	100%
	Morebayesian	100%	100%	100%	100%	100%	100%
Average		**95.69%**	**91.8%**	**93.48%**	**95.18%**	**93.6%**	**94.22%**

We employ TPR and FPR to measure the performance of detector, where TPR = Recall and FPR can be defined as follows:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{16}$$

where TN is the number of correctly detected clean examples.

The area under the receiver operating characteristic (ROC) curve (AUC) serves as a performance metric to evaluate the AE detectors.

5) Detector Setup: The method is implemented in PyTorch 1.9.0, with an input image size of 64 × 64. In the self-supervised stage, we train for 800 epochs using the Lamb optimizer, with a learning rate of 1.6×10^{-3} and the weight decay of 4×10^{-2}. We incorporate random cropping, random resizing, and random flipping into the instance binding augmentation. Subsequently, the weights of the sparse depthwise separable convolutions are transferred into the Mobilenet-V3 model, and training continues for an additional 100 epochs. For this stage, we employ the SGD optimizer, setting the momentum to 0.9 and maintaining a weight decay of 1×10^{-4}. The proposed method runs on a computer with an Intel Core i7-11700F CPU and an RTX 4090 GPU.

B. Overall Performance

1) Results on Datasets: We conduct AE detection on three datasets (DWFES, RSSCN7 and FLAME) that are subject to white-box and black-box attacks on ResNet18 and MobileNet-V3. The test images consisted of clean and AEs generated by the aforementioned nine attacks. As shown in Table II, the proposed detector achieves an average accuracy of 95.69%, an average recall of 91.80%, and an average $F1$-score of 93.48% on ResNet-18 against the adversarial attacks mentioned. Similarly, it also achieved an average accuracy of 95.18%, an average recall of 93.6%, and an average $F1$-score of 94.22% on MobileNet-V3. The proposed method demonstrated robust detection capabilities across various attacks. From the perspective of the dataset, the detection performance remains consistent across the same attack. For instance, in DWFES, ST displays the optimal detection results among all attacks, achieving a perfect score of 100% in all three metrics. Similarly, in RSSCN7, ST maintains

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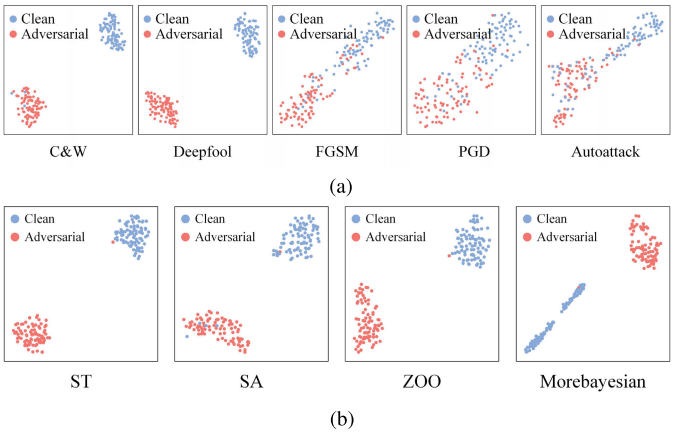


Fig. 5. Visualization of t-SNE for the white-box and black-box attacks on DWFED dataset and MobileNet-V3. Blue and red points represent feature maps corresponding to clean and AEs. (a) White-box attacks. (b) Black-box attacks.

the highest detection results among all attacks, despite slightly lower performance compared to DWFES. This can be attributed to the higher similarity between image categories in RSSCN7. From the perspective of data scale, the detection effectiveness for most attacks on the large-scale FLAME dataset outperforms that on the other two datasets. This is primarily attributed to the ample availability of clean and AE pairs in the FLAME dataset, which enables the proposed method to learn a more extensive representation of attack. From the model perspective, attacks with high detection results maintain high performance even when the attacked model is changed. Furthermore, the proposed detector consistently produces similar detection results across various models, demonstrating the transferability of the proposed detector.

Fig. 5 shows the separability of adversarial and clean examples on the DWFED dataset and MobileNet-V3 under nine considered attacks. In the proposed detector, the utilization of t-SNE for feature mapping enables the creation of a visualization chart with reduced dimensions. It can be noted that the AEs generated by the white-box attacks are more challenging to distinguish from clean examples, when compared with three black-box attacks. Given that in black-box attacks, the attacker has no knowledge of the target model, and tends to make larger perturbations to the images compared to white-box attacks. The superior performance of the proposed detector against black-box attacks compared to white-box attacks further validates this observation.

2) Performance Comparison: This article compares the proposed detector with existing state-of-the-art methods. These methods include: FS [56], LID [18], NIC [47], NSS [19], SFAD [57], and MIAED [44]. The comparison results are shown in Fig. 6. We conduct a comparison of detection performance for the nine mentioned attacks on three datasets and two attacked models. An effective detector should demonstrate both a high TPR and a low FPR. The proposed detector achieves the mentioned result, whereas some detectors only achieve high performance in one of the metrics for certain attacks, arising from the prediction bias toward particular categories. In terms of accuracy, the proposed detector achieves state-of-the-art on most attacks, demonstrating its excellent performance. On the other hand, the other detectors only demonstrate satisfactory performance against a limited number of attacks, lacking the ability to adapt to all categories of attacks. In contrast, the proposed detector achieves superior performance across all attacks. When datasets with different distributions and scales or attacked models are changed, some detectors show significant differences in the detection results for the same attacks, whereas the proposed detector maintains consistency and stability under various settings.

TABLE III
COMPARISON OF COMPLEXITY COMPARISON

Detector	MFLOPs	Params(Million)	Time(Second)
FS	6.98	2.54	0.50
LID	6.98	2.54	13.28
NIC	7.11	2.55	1.12
NSS	6.98	2.54	0.54
SFSD	9.38	2.57	15.67
MIAED	38.24	3.6	4.03
Proposed	**2.15**	**0.57**	**0.44**

In Fig. 7, the distribution and probability density are analyzed for detection accuracy to measure the comprehensive results of the detectors. Among all the violin plots, the median of the proposed detector is the highest, indicating the highest detection results. Compared to other methods, the narrower interquartile range demonstrates the consistency of the measurement value distribution. Although some methods such as FS have a narrow interquartile range, the fluctuation of the data is substantial, resulting in insufficient stability. Compared to other methods, the proposed detector can simultaneously maintain higher median results and a compact accuracy distribution. From the perspective of data scale, the results on FLAME demonstrate the consistency and stability of the proposed detector when applied to large-scale datasets.

3) Computational Complexity Comparison: Given the requirement for mobile deployment in remote sensing scenarios, the detector computational efficiency is also a measurement indicator. As shown in Table III, the proposed detector achieves floating point operations per second (FLOPs) of 2.15 MFLOPs, a total of 0.57 million parameters, and a single inference time of 0.44 s across the nine mentioned attack scenarios. The proposed method boasts low computational complexity, ensuring detection accuracy while meeting real-time performance requirements. The proposed detector achieves end-to-end outputs, leveraging sparse depthwise separable convolutions during the self-supervised training stage. During the inference stage, the weights are directly transferred to depthwise separable convolutions, thereby maintaining the computational complexity of the detector without requiring additional parameters. In contrast, other detectors obtain features through specific models and subsequently perform secondary processing on the features, and then perform detection on clean and AEs based on the analysis of the features. This two-stage detection method typically leads to an increase in the computational complexity of the detector.

C. Ablation Study

1) Effectiveness of the Proposed Module: We conduct an ablation study on the detection performance of white-box (FGSM) and black-box (ZOO) AEs generated on the RSSCN7 dataset and the attacked ResNet18 model, aiming to investigate

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Fig. 6. TPR, FPR, and accuracy results of nine attacks, the order of display in the subfigure is C&W, Deepfool, FGSM, PGD, AutoAttack, ST, SA, ZOO, and Morebayesian. Each bar chart compares the seven detection methods mentioned. (a) DWFES and ResNet18. (b) DWFES and MobileNet-V3. (c) RSSCN7 and ResNet18. (d) RSSCN7 and MobileNet-V3. (e) FLAME and ResNet18. (f) FLAME and MobileNet-V3.

the contribution of different modules. The performance is shown in Table IV, where the baseline detector employs MobileNet-V3 with additional modules removed. Furthermore, the detection results with the proposed loss are presented in No. 2 of Table IV. No. 3 of Table IV represents the detection results with the application of sparse depthwise separable convolution (SDSC). No. 4 of Table IV represents the result of applying instance binding augmentation (IBA). Finally, No. 5 displays the detection results of applying the proposed method.

It is clearly observable that in No. 1 of Table IV, the baseline's detection accuracy for both white-box and black-box data is 50%, indicating that the high similarity of AEs in remote sensing scenarios leads to the failure of the detector.

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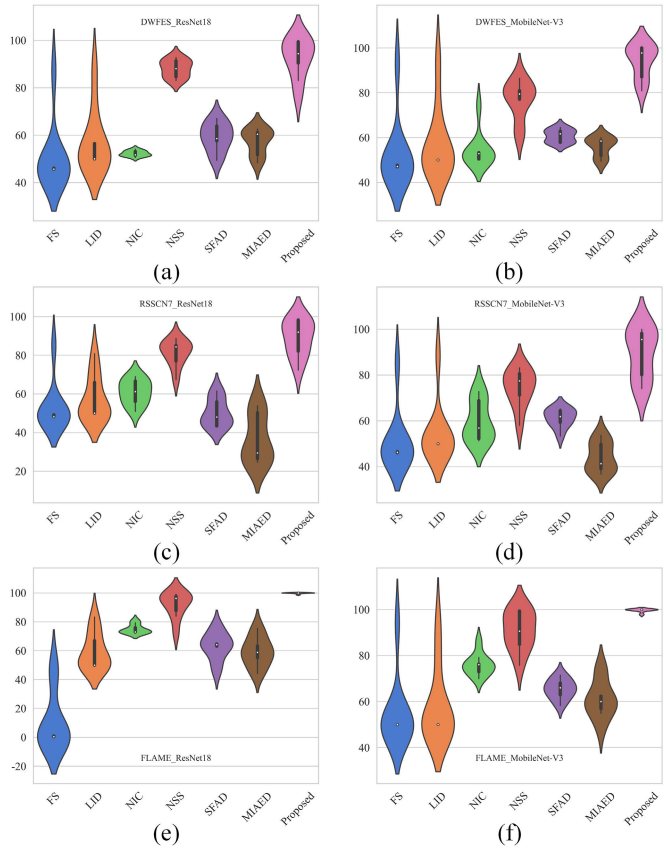


Fig. 7. Comprehensive accuracy detects results of the white-box and black-box attacks on the DWFED dataset and MobileNet-V3. (a) DWFES and ResNet18. (b) DWFES and MobileNet-V3. (c) RSSCN7 and ResNet18. (d) RSSCN7 and MobileNet-V3. (e) FLAME and ResNet18. (f) FLAME and MobileNet-V3.

TABLE IV
ABLATION EXPERIMENT OF THE PROPOSED METHOD

Variant	Candidate			White-box	Black-box
	Loss	SDEC	IBA	Acc	Acc
No.1				50.00%	50.00%
No.2	✓			72.32%	66.96%
No.3		✓		69.64%	83.04%
No.4			✓	71.43%	92.86%
No.5	✓	✓	✓	**80.14%**	**98.21%**

The results of the ablation study indicate that the selection of any one of the proposed modules can effectively detect between clean and AEs to a certain extent. Among the modules assessed, the application of instance bounding augmentation achieves a significant improvement in detection results. Specifically, compared to the baseline model, the white-box detection accuracy increased by 21.43%, while the black-box detection accuracy improved by 42.86% relative to the baseline. Notably, in the detection results of AEs generated by both black-box and white-box methods, the module's performance in black-box detection is superior to that in white-box detection, which is consistent with the mentioned observations. By applying the proposed method, classification boundaries are simultaneously strengthened and information loss is reduced, thereby achieving optimal detection performance.

TABLE V
COMPARISON OF DIFFERENT RESOLUTIONS

Resolution	Evaluation	FGSM	PGD	ST	SA	Average
64×64	TPR	73.21%	78.57%	100%	92.85%	86.15%
	FPR	8.92%	7.14%	3.57%	5.35%	6.24%
	ACC	82.14%	85.71%	98.21%	93.75%	89.95%
224×224	TPR	100%	100%	100%	96.42%	99.10%
	FPR	0	1.78%	1.78%	3.57%	1.78%
	ACC	100%	99.1%	99.1%	96.42%	98.65%
256×256	TPR	100%	100%	100%	**98.21%**	**99.55%**
	FPR	0	**0**	**0**	**0**	**0**
	ACC	100%	**100%**	**100%**	**99.1%**	**99.77%**

TABLE VI
COMPARISON OF DIFFERENT HYPERPARAMETER RATIOS

Hyperparameter rations(α/β)	White-box	Black-box
0.25	80.35%	95.53%
0.5	81.25%	96.42%
1	**83.92%**	**97.32%**
2	82.14%	95.53%
3	83.03%	94.64%

2) Effectiveness of the Resolution: This article validates the impact of adversarial images with different resolutions on the performance of the proposed detector. We conduct an ablation study setting three different resolutions of AEs ($64 \times 64, 224 \times 224, 256 \times 256$) generated on the ResNet18 based on the RSSCN7 dataset, and the results are shown in Table V. Our detector exhibits outstanding performance across three different resolutions, achieving optimal performance specifically at the 256×256. Further analysis reveals that as the resolution of the adversarial images increases, the performance of detector also improves. This amply demonstrates the positive correlation between image resolution and the clarity of attack features, suggesting that higher resolutions yield more prominent attack characteristics. With the aid of instance binding augmentation strategy, the proposed detector is able to more readily distinguish between the inherent features of the image and potential perturbations.

3) Effectiveness of the Hyperparameter Ratios: This article investigates the impact of different hyperparameter ratios ($\boldsymbol{\alpha}/\boldsymbol{\beta} = 0.25, 0.5, 1, 2, 3$) in the loss function on the performance of the proposed detector. Based on the RSSCN7 dataset, we select FGSM for white-box attacks and ZOO for black-box attacks. We evaluate these ratios on MobileNet-V3, and the results are shown in Table VI. Across all five ratios, the proposed detector exhibits excellent performance, achieving the highest performance at $\boldsymbol{\alpha}/\boldsymbol{\beta} = 1$, with an accuracy of 83.92% against FGSM and 94.64% against ZOO. This indicates that masked learning and contrastive learning achieve similar learning effects within the proposed dual-branch framework and can mutually enhance each other by updating convolutional kernel parameters.

4) Generalization of the Proposed Detector: This article further assesses the adversarial generalization ability of the proposed detector under various conditions to study the robustness of the method against unknown attacks. The experimental setup is as follows. The experimental setup involves initially training the detector on AEs generated by two specific attack

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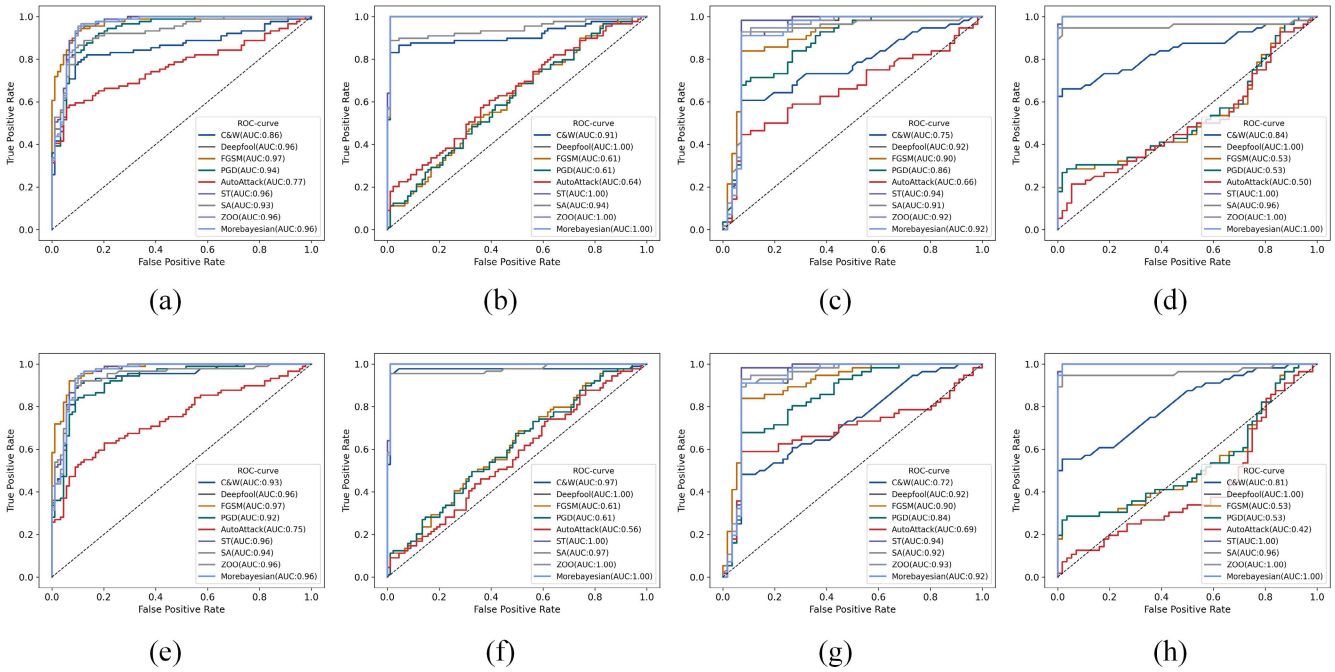


Fig. 8. Generalization experiments conduct across various datasets and attacked models evaluate the ROC curves and AUC values of nine different attacks under specific detectors, with each subfigure representing a particular combination of configurations. The white-box detector and black-box detector are trained on DWFES and ResNet18 using AEs generated by FGSM and ZOO, respectively. (a) DWFES, ResNet18 and white-box. (b) DWFES, ResNet18 and black-box. (c) RSSCN7, ResNet18 and white-box. (d) RSSCN7, ResNet18, and black-box. (e) DWFES, MobileNet-V3, and white-box. (f) DWFES, MobileNet-V3, and black-box. (g) RSSCN7, MobileNet-V3, and white-box. (h) RSSCN7, MobileNet-V3, and black-box.

methods: FGSM and ZOO. These AEs are crafted using a ResNet18 model trained on the DWFES dataset. After being trained on AEs generated by the known FGSM and ZOO attacks, the detector is then employed to perform inference on AEs crafted by eight unknown attack methods, utilizing diverse datasets and targeting different attacked models. It is important to note that both the white-box detector trained specifically on FGSM-generated AEs and the black-box detector trained on ZOO-generated AEs possess no prior information regarding the aforementioned unknown AEs utilized during the evaluation stage. As shown in Fig. 8, the proposed detector demonstrates remarkable generalization capabilities across various scenarios, models, and attacks, achieving an AUC of over 90% in the majority of unknown AEs.

In Fig. 8(a) and (b), it can be seen that the white-box detector (FGSM) achieves superior stability compared to the black-box detector (ZOO). The black-box detector performs poorly against some attacks, which is attributable to the lower complexity of features associated with black-box attacks. The black-box detector achieves poor performance against certain attacks, attributed to the lower complexity of features, which leads to a reduction in the representational ability of the detector. In Fig. 8(c) and (d), the dataset is replaced from DWFES to RSSCN7 while maintaining the application of the two detectors. Notably, both the black-box and white-box detectors experience a certain decrease in AUC. Nevertheless, detectors still maintain competitive results. In Fig. 8(e) and (f), the attacked model is replaced from ResNet18 to MobileNet-V3. Notably, some detection results even achieve an increase in AUC. This improvement can be attributed to the higher complexity of ResNet18 compared to MobileNet-V3, which results in the detector providing stronger detection capabilities. In Fig. 8(g) and (h), the dataset and the attacked model are simultaneously changed. It is noteworthy that, while the AUC remains similar after changing the attacked model, the detection performance is the worst among all conditions when both the dataset and the model are changed. By leveraging the similarity factors among remote sensing adversarial images, models, and attacks, we can deploy a specific attack detector on mobile devices, ensuring effective detection of known adversarial attacks while maintaining a certain ability to detect unknown attacks.

V. Conclusion

In this study, we confront the challenge of AE detection in remote sensing scenarios on mobile devices by proposing a contrastive masked detection method. The framework consists of a dual-branch structure to simultaneously reconstruct perturbation distributions and facilitate both intra-instance and inter-example feature discrimination. The proposed method enhances the classification boundary between adversarial and clean examples by applying a instance binding augmentation strategy. Given that deploying on mobile devices, our method employs a sparse hierarchical encode modeling without increasing additional parameters. Furthermore, we conduct extensive experiments on both white-box and black-box AEs, demonstrating that our proposed method significantly outperforms other state-of-the-art methods and displays defense characteristics against unknown adversarial attacks. Although our method demonstrates robust performance in

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adversarial detection, it remains a challenge to detect the optimal combination of attack strategies and the precise sequence of attacks when an attacker employs multiple attacks on the same remote sensing image. In future research, we will incorporate multilabel learning into our method to further delve into the formation process of attacked images.

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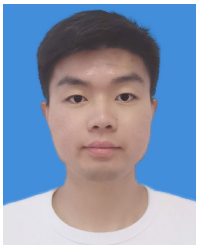
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